



NeoPixel Colour-Changing LED Lighting Using Arduino

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This NeoPixel LED lighting based on WS2811 driver and Arduino digitally produces red, green, and blue (RGB) lights and the combinations. Originally introduced by Adafruit, the LED strips come with driver ICs that are addressable and programmable.

There are several types of LED strips/arrays available but we may

use the one based on WS2811 driver chip. However, its equivalents WS2812, WS2812b, and WS2813 can also be used. Serial protocol bus used to address and control the LEDs can be somewhat complex and tricky as the drivers can generate quite a few lighting effects and patterns like zig zag, random, 7-segment digital clock, scrolling message displays,

etc. The author's prototype is shown in Fig. 1.

The NeoPixel LED lighting available in the market is usually costly. The use of WS2811 driver in this project helps reduce cost. NeoPixel LEDs come in a wide variety of pixel densities and form factors, besides the traditional LED strip, such as rings, matrices, and several other types of individual pixels as shown in Fig. 2.

Controlling a WS2811 device via its serial control bus is not a straightforward task. It uses a combined clock and data line to ripple the instructions for each LED down the serial daisy chain bus, which runs from one pixel to the next (Fig. 3). This means the pixels can not be addressed individually. So, any change in one of the pixels would require refreshing the instructions loaded into all the pixels upstream. RGB LED's pin diagrams are shown in Fig. 4.

Controlling NeoPixel LEDs

Generating instructions for a string of WS2811 LED drivers requires very specific and complicated timing that can make the driver code a bit tricky to write from scratch. Fortunately, much of the hard work has already been done and there are libraries (FastLed.h, NeoPixel.h) available for LED strip drivers and applications available for Arduino,

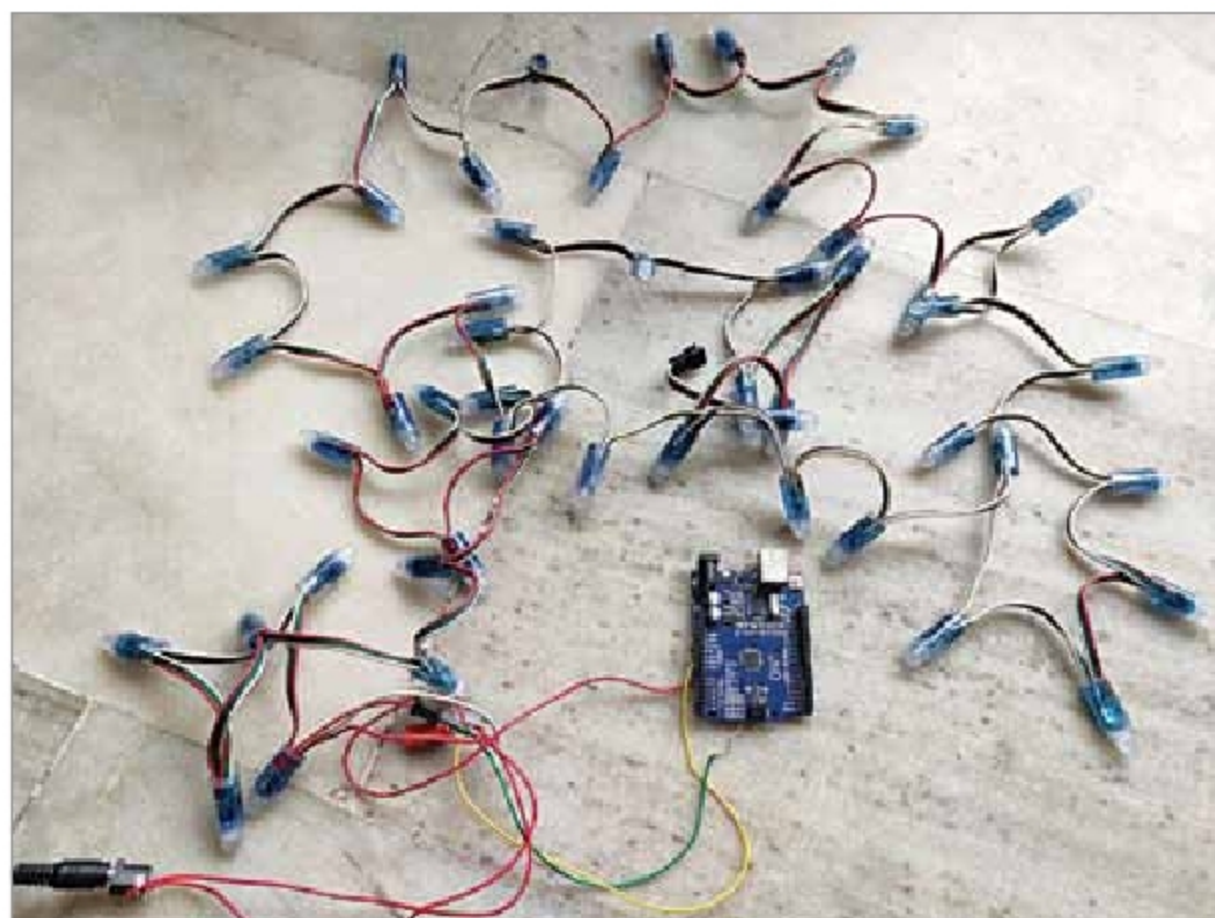


Fig. 1: Author's prototype

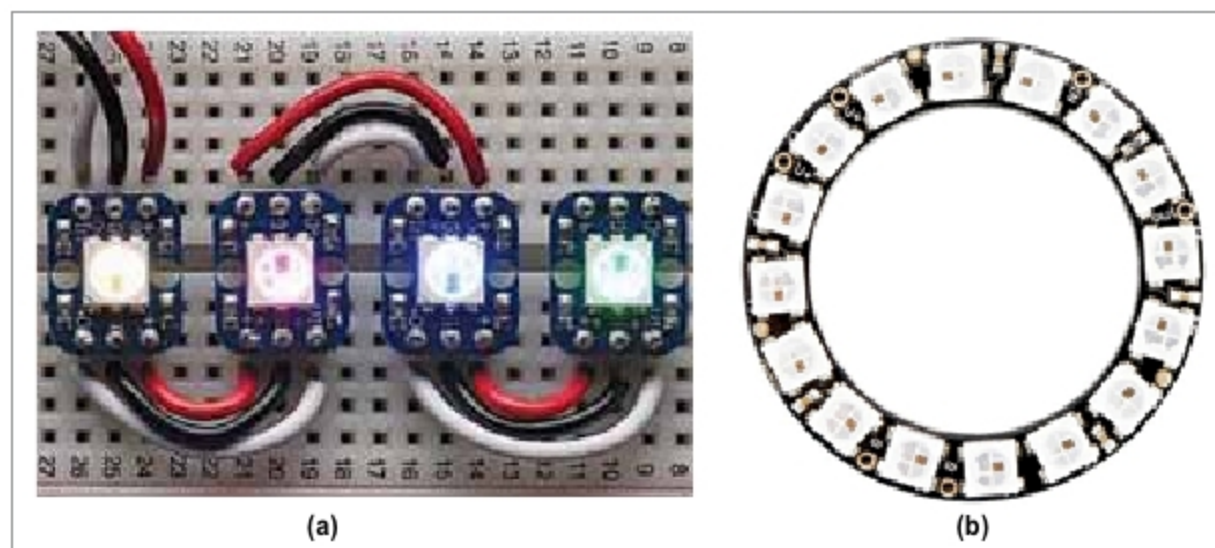


Fig. 2: Two types of NeoPixel LED strings: (a) matrix type, (b) ring type

PARTS LIST

- Semiconductors:**
- Arduino Uno
 - 50-LED NeoPixel strip
- Resistors (all 1/4-watt, $\pm 5\%$ carbon):**
- 470-ohm
- Capacitors:**
- 1000 μ F, 16V electrolytic
- Miscellaneous:**
- 5V, 5A SMPS DC adaptor
 - Jumper wires